Accelerating Hyperspectral Anomaly Detection With Enhanced Multivariate Gaussianization Based on FPGA

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***Abstract*— Hyperspectral anomaly detection (AD), as a frontier research topic in the field of remotely sensed data processing, aims to identify targets of interest from complex and vast images. Existing AD methods typically involve complex models and many parameters, posing challenges in meeting the requirements of computational efficiency in hyperspectral AD. To address this issue, this article presents an AD acceleration algorithm based on the multivariate Gaussian model as well as its field programmable gate array (FPGA) implementation. By exploiting the parallel processing capabilities of FPGA, we introduce an innovative spectral dimensionality reduction method in which the data processing flow can be accomplished in a distributed manner. Then, we employ an improved linear rotation strategy based on correlation coefficients to accelerate the convergence rate of the proposed AD algorithm. The rotation of Gaussianization in the improved strategy is independent of eigenvalue decomposition, thereby substantially reducing the computational complexity involved during the rotation procedure. Furthermore, we apply a pipeline parallel mechanism to facilitate the FPGA implementation of the AD algorithm and to significantly enhance the computational efficiency. Experimental results on an embedded FPGA platform demonstrate that the FPGA implementation of the hyperspectral AD algorithm proposed in this article achieves a significant acceleration rate with guaranteed high detection accuracy.**

***Index Terms*— Anomaly detection (AD), field programmable gate array (FPGA), hyperspectral image (HSI), linear rotation.**

I. Introduction

HYPERSPECTRAL images (HSIs) are a type of 3-D data collected by imaging spectrometers, containing not only spatial information but also rich spectral information. Their spectral resolution can reach the nanometer level, and the continuous curves can precisely reveal the material properties inherent in the spectrum, providing guidance and support for terrain classification and target detection. Anomaly detection (AD) is an important research direction in hyperspectral data processing. Compared to target detection tasks, AD does not require prior spectral information of the targets, which holds greater application value in complex geographical environments [1]. To date, hyperspectral AD has been widely applied, showcasing its unique advantages in various military and civilian fields, such as revealing camouflage, identifying small targets, and determining land categories [2], [3], [4].

In the literature, the hyperspectral AD methods can be mainly classified into three categories: statistical methods, representation-based methods, and deep learning-based methods. Statistical models, as a classic AD approach, are based on the core assumption that background data follows a specific statistical distribution, while anomalous data deviate from this pattern. The Reed–Xiaoli (RX) algorithm, proposed in 1990 [5], is the most widely used statistical model method and has become a benchmark for AD methods. RX performs statistical analysis of the background distribution using the entire image and then uses the Mahalanobis distance to measure the deviation between the pixel to be detected and the background. Many improved variants of the RX algorithm have emerged. Local RX (LRX) [6] uses data within a local window to compute the covariance matrix, considering that within a smaller range, the distribution of features in HSIs is more uniform, and the data consistency is stronger, making it more accurate to describe the data with a Gaussian distribution. Another RX variant introduced kernel methods into the RX detector, which significantly improved its performance [7]. Kernel RX extends the traditional RX algorithm to a higher dimensional feature space, turning the nonlinear problems in the original space that are difficult to solve into linear problems in the new high-dimensional space for effective processing. This simplification aids in AD and bolsters the algorithm's capacity to handle intricate hyperspectral data.

Representation-based methods constitute another class of AD methods that have gained considerable attention. The fundamental principle of these methods involves representing data as a linear combination of a set of base elements, which can be learned from the original data [8], [9], [10]. For example, Candès et al. [11] proposed a method based on the theory of sparse representation, where each pixel of the image is represented by a linear combination of dictionary atoms,

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and residual analysis is performed between the original and represented images to identify anomalous targets within the image. In the analysis of HSIs, considering that pixels within a local area often represent the same type of material, each pixel can be represented by its neighboring pixels. Building on this concept, Li and Du [12] introduced a collaborative representation-based detector (CRD) for AD. CRD relies on precise background features and powerful image representation capabilities, which pose significant challenges in practical applications. To overcome this challenge, Chang and Ghamisi [13] further proposed nonnegative-constrained joint collaborative representation (NJCR), which, by constructing a joint dictionary composed of background and anomaly subdictionaries, effectively extracts reliable samples. With the swift advancements in artificial intelligence (AI), deep learning methods have become increasingly prevalent in remote sensing. For instance, Hong et al. [14] developed a large-scale model tailored for hyperspectral remote sensing data, boasting over 600 million parameters. Huyan et al. [15] presented a few-shot learning AD network that employs relational learning to adapt the network to previously unseen datasets. Li et al. [16] proposed a low-rank representation network (LRR-Net) to enhance AD performance by incorporating the LRR model as prior knowledge into the deep learning network to guide parameter optimization. Moreover, they introduced a method for learning disentangled priors [17], which captures complete prior knowledge to effectively model the background, thereby extracting anomalies from HSIs more accurately.

Compared to statistical AD methods, representation-based methods require complex matrix computations, while deep learning methods are often resource-intensive in terms of interpretation. In recent studies, researchers have developed other statistical methods that achieve excellent detection performance. Yang et al. [18], the stochastic RX detector method was employed, which conducts AD in HSIs for a variety of features, such as morphological and Gabor features, significantly enhancing the reliability of AD through the application of ensemble learning principles. Beyond RX-based algorithms, Chang [19] introduced the constrained energy minimization (CEM) detector into AD, where the main idea is to transform known specific targets applied on CEM into unknown specific targets. Results show that the CEM detector also exhibits outstanding performance in AD tasks. Padrón-Hidalgo et al. [20] proposed rotation-based iterative Gaussianization (RBIG) for AD using Gaussianization to estimate the probability density distribution of each pixel in the image. However, due to line rotation operations, RBIG may suffer from a slow convergence rate and high-computational complexity.

In pursuit of higher detection accuracy, many algorithms tend to increase the complexity of the model. Although the increase may improve detection performance, it also increases computational costs. Such high-computational costs are often unacceptable in practical applications, especially in time-sensitive military and security fields. To enhance the speed and performance of AD, field programmable gate arrays (FPGAs) have emerged as a favorable platform for algorithm development. Their inherent capability for pipelined processing and superior computational power, when compared to central processing units (CPUs), make FPGAs an attractive option. However, the current landscape of FPGA-specific AD algorithms remains sparse, primarily due to the lengthy development cycles and the intricacies associated with hardware-level coding. Furthermore, the majority of existing algorithms deployed on FPGAs for AD, such as RX and automatic target detection and classification algorithms (ATDCAs), tend to be relatively simplistic in nature. These algorithms often fail to fully exploit the rich spatial and spectral information present in HSIs, resulting in suboptimal detection outcomes that are less than satisfactory.

With the above-mentioned two concerns in mind, we propose optimized RBIG (ORBIG), an FPGA-optimized algorithm tailored specifically for hyperspectral AD. Distinct from the conventional RBIG, our proposed ORBIG meticulously incorporates the comprehensive data information present in HSIs, while simultaneously accounting for the unique hardware characteristics of FPGAs. This approach introduces Gaussianization to the scientific advancement of FPGA-based HSI processing, adhering to the rigorous standards expected in academic publications. First, considering the coding characteristics of FPGAs and the sparse nature of anomalies in HSIs, we introduce average fusion to replace traditional feature extraction techniques. This fusion approach enhances parallelism and better preserves anomaly characteristics compared to existing methods like principal component analysis (PCA). Subsequently, this article proposes an innovative rotation iterative method based on correlation coefficients, specifically designed to address the high-computational complexity and slow convergence issues present in existing rotation methods like PCA and random rotations (RND) [21]. This method avoids eigenvector computations by using the first few columns with the lowest correlation for orthogonalization calculations, reducing computational complexity and mitigating the slow convergence issue associated with the randomness of orthogonal matrices in the RND method. Furthermore, the article optimizes the calculation of the inverse cumulative distribution function (cdf) using a binary search method, reducing complexity and making it better suited for FPGA development. Finally, this article achieves parallel optimization of the algorithm on FPGA through pipeline computation, significantly enhancing its efficiency. Overall, this work contributes to the following.

1) We propose an optimized Gaussian-based hyperspectral AD method that identifies anomaly locations by estimating the data probability distribution.
2) We propose a new rotation method based on correlation coefficients, to avoid complex computations and slow convergence in the existing RBIG method.
3) We optimize the calculation of the proposed algorithm to facilitate its distributed and parallel implementation on FPGA.

The remainder of this article is organized as follows: Section II introduces the related work on accelerating computation. Section III provides a detailed introduction to the hyperspectral AD method based on Gaussian rotation iteration. Section IV presents an optimized version of RBIG. Section V

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elaborates on the implementation of the ORBIG algorithm on FPGA. Section VI demonstrates the reliability and acceleration effect of the method through experimental evidence. Finally, Section VII concludes the article.

II. Related Work

A. AD Based on Distributed Platform

Optimizing the algorithm structure and employing parallel computing of data can significantly improve the operational efficiency of algorithms. For example, Zhang et al. [22] have introduced a distributed and parallel low-rank and sparse representation (DP-LRASR) computational method for HSIs, achieving efficient parallel detection. Díaz, et al. [23] developed an AD algorithm based on orthogonal projection techniques, which, by adopting a streamlined and highly parallel computation process, effectively avoids the need for complex matrix operations. These innovations optimize the computational process and significantly increase data processing speed through parallel computing technologies, offering a viable technical pathway for real-time HSI analysis and AD.

B. AD Based on FPGA

HSI processing methods optimized for hardware have gained widespread recognition, especially computations based on FPGAs. As a hardware acceleration device, FPGA offers high flexibility, strong processing capabilities, and compact size, making them highly suitable for onboard or satellite computing needs. Detection algorithms developed with FPGAs can meet the timeliness requirements of detection tasks well and are better suited for complex application scenarios. Lei et al. [24] proposed a fast spectral–spatial AD algorithm based on morphological reconstruction and simplified guided filtering (FAST-MGD). The algorithm processes images using morphological reconstruction and a simplified guided filter and utilizes a deep pipeline for acceleration to achieve better time and accuracy results. Díaz et al. [25] proposed an FPGA-based AD algorithm for push-broom hyperspectral imaging systems, utilizing an orthogonal projection strategy to estimate the orthogonal subspace spanned by the background distribution, within which anomalies can be more effectively detected. For this purpose, image pixel blocks are processed independently, eliminating spatial alignment constraints. The FPGA-based hyperspectral AD algorithm features high performance, low-power consumption, parallel processing capabilities, and flexibility, allowing for real-time or near-real-time data processing and AD. It is suitable for fast, efficient, and customized AD tasks in hyperspectral data. However, algorithm development and optimization require substantial hardware design and programming knowledge, with careful attention to resource utilization, efficiency, and balance.

HSI, with its high spectral resolution and rich spatial–spectral information, demonstrates immense application potential in both military and civilian domains. However, the complexity of HSIs, especially their massive data volume and numerous spectral bands, poses challenges for data processing [26], [27]. This requires specialized algorithms and techniques for data dimensionality reduction, feature extraction, and AD to analyze HSIs effectively.

III. Algorithm

In this section, we introduce the principle of the RBIG, starting with an overview of its core concept and then explaining its application in hyperspectral AD.

A. RBIG

The multivariate Gaussian detection method discussed in this article aims to identify a transformation that maps a dataset into a space conforming to a multivariate Gaussian distribution, thereby enabling the estimation of its multivariate distribution density. Specifically, for a dataset $\boldsymbol{Y} \in \mathbb{R}^{l\times d}$, RBIG seeks to find a transformation $\boldsymbol{G}_y$ that maps the data $\boldsymbol{Y}$ into the space of a new dataset $\boldsymbol{X} \in \mathbb{R}^{l\times d}$, where $\boldsymbol{X}$ follows a multivariate Gaussian distribution:

$$\begin{aligned}\boldsymbol{G}_y : \boldsymbol{y} \in \mathbb{R}^d \rightarrow \boldsymbol{x} \in \mathbb{R}^d \\ \sim p_Y(\boldsymbol{y}) p_X(\boldsymbol{x}) \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}_d).\end{aligned} \tag{1}$$

The input data points $\boldsymbol{y} \in \mathbb{R}^d$ share the same data dimensionality with map data $\boldsymbol{x} \in \mathbb{R}^d$, l is the number of atoms, $\boldsymbol{0}$ represents a zero vector, and $\boldsymbol{I}_d$ denoting the identity matrix. Transforming the formula allows for estimating the probability of a point $\boldsymbol{y}$ in the original domain

$$p_Y(\boldsymbol{y}) = p_X(\boldsymbol{x})\left|\boldsymbol{J}_{\boldsymbol{G}_y}(\boldsymbol{x})\right| \tag{2}$$

where $p_Y(\boldsymbol{y})$ represents the probability distribution of the original data point $\boldsymbol{y}$, and $p_X(\boldsymbol{x})$ indicates the probability density of the data point after transformation in Gaussian space. It is known that p_X follows a Gaussian distribution with a mean of $\boldsymbol{0}$ and a covariance matrix as the identity matrix, and $|\boldsymbol{J}_{\boldsymbol{G}_y}(\boldsymbol{x})|$ represents the Jacobian determinant obtained at point $\boldsymbol{x}$ through the transformation $\boldsymbol{G}_y$. The Jacobian determinant refers to the scaling factor of volume between coordinate systems before and after the transformation, measuring the impact of the transformation on volume. Here, $|\boldsymbol{J}_{\boldsymbol{G}_y}(\boldsymbol{x})|$ measures the local linear scaling effect of the transformation process from data point $\boldsymbol{y}$ to $\boldsymbol{x}$.

From a computational perspective, RBIG is an iterative process, with each iteration primarily comprising two steps: 1) a series of marginal Gaussianization operations $\boldsymbol{\psi}$ are performed on the variables and 2) a linear rotation $\boldsymbol{R} \in \mathbb{R}^{d\times d}$ is conducted. The equation is

$$\boldsymbol{y}_{i+1} = \boldsymbol{R}_i \cdot \boldsymbol{\psi}_i(\boldsymbol{y}_i), \quad i = 1, \ldots, \boldsymbol{M} \tag{3}$$

and $\boldsymbol{M}$ represents the number of iterative steps, where $i = 1, \ldots, \boldsymbol{M}$. The process of marginal Gaussianization for each dimension can be decomposed into two equalizing transformations. First, marginal uniformization is carried out based on the cdf of the marginal probability density function (pdf). Second, uniform variables are transformed into Gaussian-distributed variables based on the inverse function of the univariate Gaussian cumulative density function. As demonstrated in [21], with a sufficient number of iterations, the transformation will ultimately converge, and the resulting data will adhere to a

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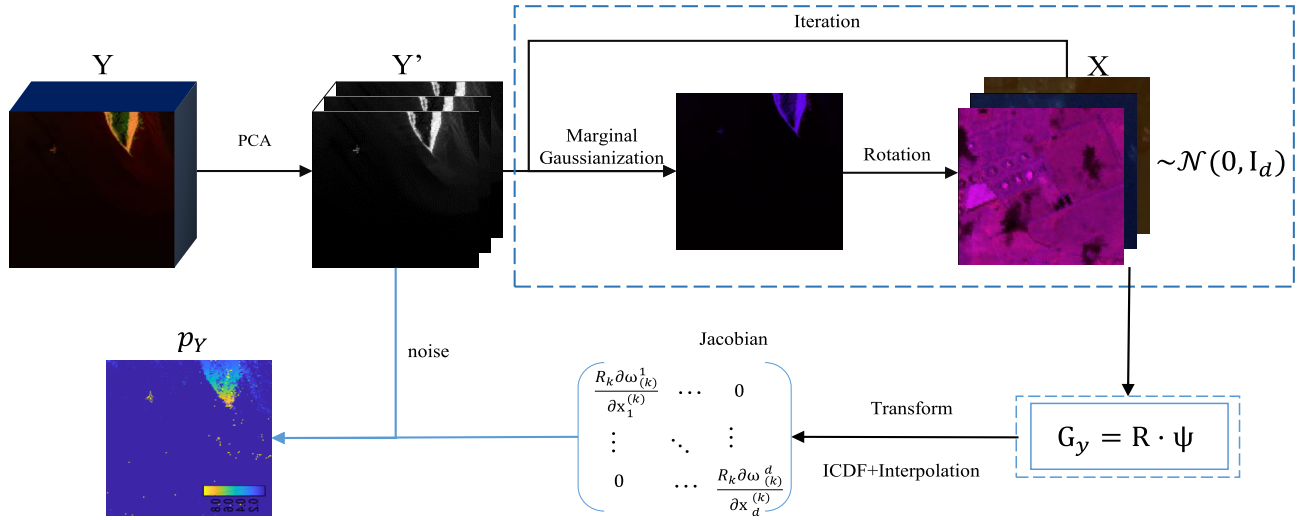


Fig. 1. RBIG algorithm process.

standard Gaussian distribution. In applying this method, the calculation of the Jacobian determinant depends solely on the number of iterations. It is important to note that since the linear transformation involves only rotation operations, its Jacobian determinant consistently remains equal to 1. Currently, the primary rotation iterative methods for data are PCA rotation and RND. PCA relies on eigenvalue decomposition of the original data, offering fast convergence but high computational complexity. The RND method, on the other hand, employs random numbers and orthogonal vectors for transformation, resulting in slower convergence. To address these issues, this article introduces a new rotation method $\boldsymbol{R}$, which optimizes computation while ensuring improved convergence speed for data transformation. The specific implementation process will be detailed in Section IV.

B. RBIG Applied to AD

Through a deep understanding of the RX algorithm, we can recognize the applicative value of RBIG in AD. The principle behind the RX algorithm is that the lower the probability of a data point's occurrence, the higher the likelihood of it being an anomaly [28]. Although the RX algorithm is widely adopted due to its intuitiveness and effectiveness, it also faces some limitations, primarily in its limited capacity to process high-dimensional data and potentially high false positive rates under certain conditions. To address these limitations of the RX algorithm, this article proposes the RBIG method to estimate the pdf as a more direct and effective means. RBIG, an unsupervised learning method, has the advantage of not requiring labeled data for training, nor does it necessitate preprocessing steps such as linear data scaling. This study utilizes RBIG to calculate probability distributions, not only effectively identifying anomalies but also enhancing the accuracy and robustness of AD. The model is as follows:

$$\boldsymbol{A}(\boldsymbol{y}) \propto \frac{1}{\boldsymbol{p}_Y(\boldsymbol{y})}. \tag{4}$$

The flowchart of RBIG is shown in Fig. 1.

Hyperspectral data typically consists of images across multiple wavelengths, each corresponding to a spectral band. Therefore, the dataset $\boldsymbol{Y}$ is initially subjected to dimensionality reduction to obtain $\boldsymbol{Y'}$, reducing noise and extracting more compelling features. Subsequently, the data undergo a crucial iterative process where marginal Gaussianization aims to transform each dimension of the hyperspectral data into a form that conforms to a Gaussian distribution. At the same time, the rotation step reveals its inherent structure and patterns. This series of operations maps the data to a new data space, where the data exhibit a standard multivariate normal distribution. The Jacobian determinant is used to calculate the probability density of the transformed data. Ultimately, through the computation, the probability density $\boldsymbol{p}_Y(\boldsymbol{y})$ of the data points is obtained, where the probability of anomalies is relatively low, and that of background pixels is relatively high. In addition to the aforementioned rotation process, we optimize the dimensionality reduction method by using average fusion instead of PCA. This approach better preserves the intrinsic nature of anomaly information while also reducing computational complexity.

IV. Optimized Rotation-Based Iterative Gaussianization

In this section, we will elaborate on the optimization process of the ORBIG algorithm. As depicted in Fig. 2, the optimized ORBIG algorithm process begins with the dimensionality reduction of the HSI through band-averaging fusion. Next, the spectral matrix undergoes marginal Gaussianization to normalize the data distribution. The Gaussianized results are then rotated based on correlation coefficients to further align the data. Subsequently, the transformed results are used to obtain the probability density of the image in the original domain. This involves transforming the data using a Jacobian matrix, applying a binary inverse cdf with interpolation, and ultimately generating the pdf. The final step involves calculating the detection results from the probability density to identify anomalies, leveraging advanced data transformation

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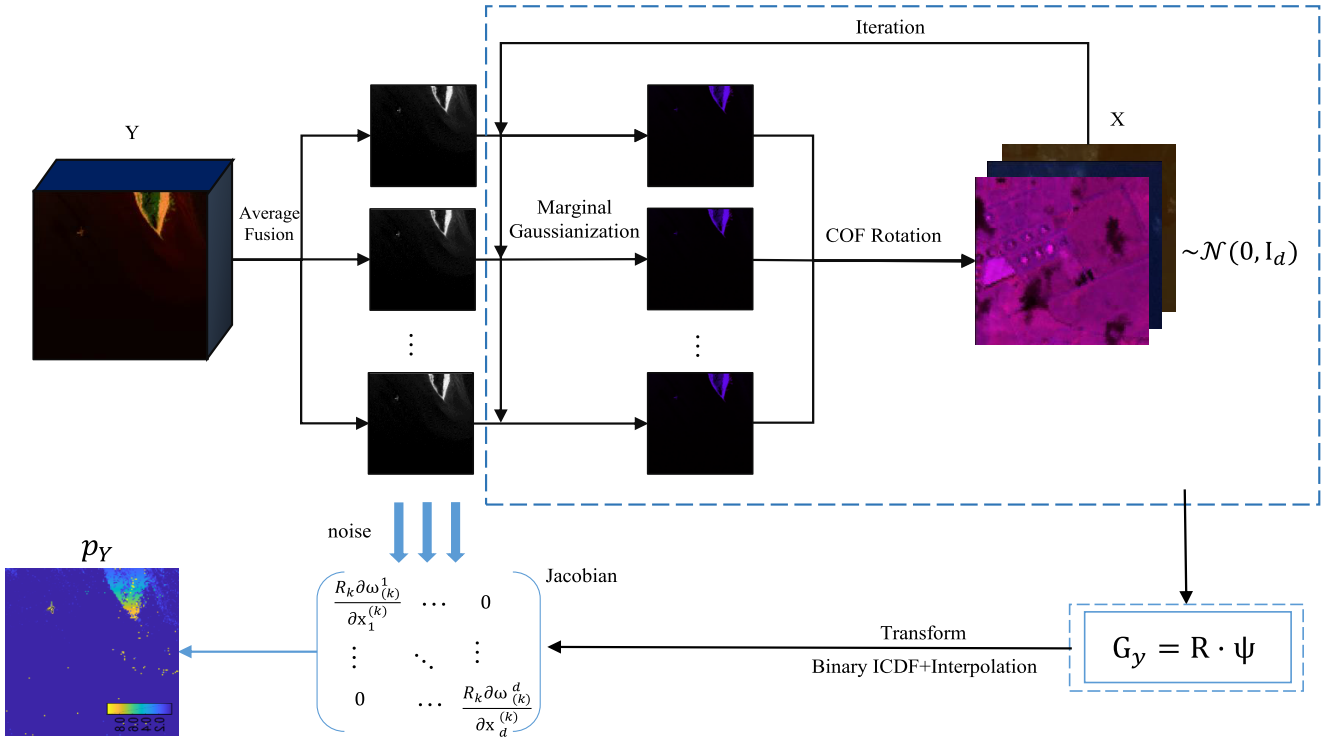


Fig. 2. ORBIG algorithm process.

and normalization techniques to enhance the accuracy and reliability of hyperspectral AD.

A. Average Fusion

The high-dimensional nature of HSIs poses a significant computational burden and demands substantial storage space. In fact, by calculating the correlation coefficients between spectral curves, it can be observed that adjacent spectral curves often exhibit significant correlations, indicating a large amount of redundant information. Therefore, the correlation coefficients between adjacent spectral bands are an important indicator for measuring this redundancy. The calculation of correlation coefficients can determine the strength of the linear relationships between spectral bands, thereby providing a theoretical basis for dimensionality reduction strategies and data compression

$$\rho_{i,j} = \frac{\mathrm{Cov}(\boldsymbol{g}_i, \boldsymbol{g}_j)}{\sqrt{\mathrm{Var}(\boldsymbol{g}_i)\mathrm{Var}(\boldsymbol{g}_j)}} \tag{5}$$

where Cov and Var represent covariance and variance, respectively, $\boldsymbol{g}_i$ and $\boldsymbol{g}_j$ refer to the ith and jth hyperspectral bands. Calculating the correlation among the spectral of HSIs yields the following results. As shown in Fig. 3, the observed correlation coefficients between adjacent bands can reach as high as 0.9997, and the lowest is 0.8871. It is evident that by fusing the data of adjacent bands, we can effectively reduce the dimensionality of the data while still retaining the key characteristics of the original data space, ensuring that important feature information is not lost. Furthermore, band fusion helps to smooth random noise because it reduces the impact of noise in a single band by integrating data

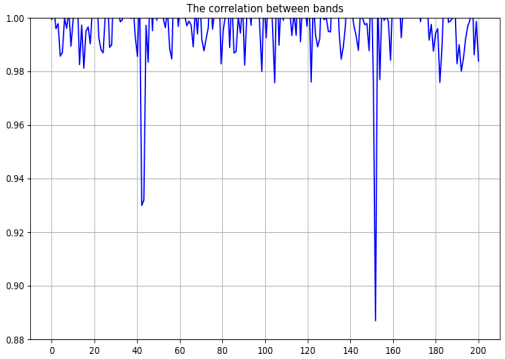


Fig. 3. Curve of correlation coefficients of adjacent spectral bands.

from multiple bands, thereby effectively enhancing the signal-to-noise ratio. Selecting and combining those representative bands with higher correlations can strengthen the key signals in the data, which is of significant value for distinguishing between anomalies and background information. Based on the above expression, the HSI is first divided into $\boldsymbol{q}$ subsets of adjacent bands

$$\boldsymbol{C}^k = \begin{cases} (\boldsymbol{g}_{\lceil d/q \rceil (k-1)+1}, \ldots, \boldsymbol{g}_{\lceil d/q \rceil k}), & 1 \le k \le q-1 \\ (\boldsymbol{g}_{\lceil d/q \rceil (k-1)+1}, \ldots, \boldsymbol{g}_d), & k = q. \end{cases} \tag{6}$$

In the above formula, $\boldsymbol{C}^k$ represents the kth hyperspectral subset, and $\boldsymbol{g} = (\boldsymbol{g}_1, \boldsymbol{g}_2, \ldots, \boldsymbol{g}_d)$ denotes the HSI having d bands in the spatial domain. $\lceil d/q \rceil$ denotes the smallest integer greater than or equal to d/q. Then, the band dimension is reduced by employing the average fusion method [29], with

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the specific calculation as follows:

$$\boldsymbol{S}^k = \frac{\sum_{i=1}^{\boldsymbol{N}_k} \boldsymbol{C}_i^k}{\boldsymbol{N}_k}. \tag{7}$$

$\boldsymbol{C}_i^k$ represents the ith band in the kth hyperspectral subset, and $\boldsymbol{N}_k$ is the total number of spectral bands in the kth subset. Applying dimensionality reduction techniques helps prevent overfitting, enhancing the model's generalization performance on unknown datasets. Furthermore, dimensionality reduction makes the data more concise, facilitating in-depth analysis and intuitive visualization. Implementing band fusion as a means of dimensionality reduction on FPGA platforms has a clear computational advantage over other methods. Band fusion primarily involves simple addition and division operations, with significantly lower computational complexity than statistical dimensionality reduction techniques like PCA, which usually require more complex matrix operations and eigendecomposition. Additionally, a key advantage of band fusion is its lower demand for data, which does not require the global information of the entire dataset but allows operation on selected data subsets only. This method is particularly suitable for FPGA environments with limited storage resources, as it allows the limited memory resources to be used to store the currently processed subset, thereby making more efficient allocation and utilization of computational resources possible. This processing strategy optimizes memory use and improves the throughput of data processing while maintaining the real-time nature of the process, which is especially important for applications requiring rapid data processing.

B. Correlation Coefficient-Based Iterative Rotation Method

In the ORBIG framework, the rotation process targets data that have already undergone marginal Gaussianization, and adopting different rotation strategies will affect the characteristics of the algorithm. Under the premise of ensuring the Gaussianization process can converge, the key criterion for selecting the rotation matrix lies in pursuing a better convergence speed. Taking PCA and RND as examples, their rotation processes are specifically described as follows.

For the PCA rotation.

1) *Covariance Matrix Calculation:* Calculate the covariance matrix of the sample data.
2) *Eigenvalue Decomposition:* Perform eigenvalue decomposition on the covariance matrix to obtain eigenvalues and corresponding eigenvectors $\boldsymbol{V}$.
3) *Data Transformation:* Perform matrix multiplication of the sample data with the obtained eigenvectors $\boldsymbol{V}$ to project the data into the space formed by the principal components.

For the RND method.

1) *Initialization:* Generate a random matrix $\boldsymbol{V}$.
2) *Orthogonalization:* Orthogonalize matrix $\boldsymbol{V}$ to ensure statistical independence of its columns.
3) *Normalization:* Normalize the orthogonalized matrix so that its columns have unit norm.
4) *Data Transformation:* Perform matrix multiplication of the marginal Gaussianization sample data with the orthogonalized and normalized matrix.

Using PCA rotation can effectively eliminate the second-order redundancy in data, but it does not maximize the reduction of marginal non-Gaussianity. The orthogonal matrix upon which PCA rotation relies is obtained based on the eigenvalue decomposition of the data, a process relatively complex in hardware programming, making PCA not the best choice from the perspective of development and computational efficiency. In contrast, the RND transformation theoretically ensures the method's convergence by guaranteeing that the algorithm does not converge to a non-Gaussian solution, even in specific singular scenarios. If after an infinite number of RND, the algorithm achieves marginal non-Gaussianity reduced to zero, it indicates that the desired Gaussian solution has ultimately been achieved. Since the RND matrix does not depend on rotation data, its convergence process may be more costly. In practical applications, the rotation matrix can be obtained through the orthogonalization process to optimize cost and efficiency based on the few columns of sample data with the lowest correlation coefficients. This method can achieve better cost-effectiveness while ensuring theoretical convergence. The process is in Algorithm 1.

Algorithm 1 Rotation Based on Correlation

Require: data samples $\boldsymbol{Y}$, bands d

1: calculate column correlations $correlation$
2: select the first d columns with the minimum correlation
$cor = \text{least}\ (correlation, d)$
3: choose the corresponding columns of $\boldsymbol{Y}$ for $\boldsymbol{V}$
$\boldsymbol{V} = \boldsymbol{Y}(cor)$
4: Orthogonalize and normalize $\boldsymbol{V}$
$\boldsymbol{V} = \text{ort}\ (\boldsymbol{V})$, $\boldsymbol{V} = \text{nor}\ (\boldsymbol{V})$
5: $\boldsymbol{Y} = \boldsymbol{Y} \cdot \boldsymbol{V}$
6: **return** $\boldsymbol{Y}$

The optimization of the rotation process in the RBIG framework aims to optimize the convergence speed and computational efficiency of the algorithm by judiciously choosing the rotation strategy. In specific computations, it is not necessary to calculate the correlation between all columns; it suffices to select one column and compute its correlation with others, thereby reducing the scale of the correlation matrix from $\boldsymbol{N}^2$ to $\boldsymbol{N}$, this reduction lowers the complexity and enhances the cost-effectiveness of programming in hardware implementations. Therefore, the rotation method based on correlation coefficients is a better choice.

V. FPGA Implementation

FPGA is a highly flexible and configurable hardware platform that allows developers to change its functionality through reprogramming and adapting to various applications and requirements. A notable advantage of FPGA is its parallel processing capability, allowing simultaneous execution of multiple operations, significantly enhancing performance for applications requiring extensive parallel computation [30]. Furthermore, FPGAs offer lower processing latency than CPUs and graphics processing units (GPUs), which is crucial for real-time systems and time-sensitive applications. Not only

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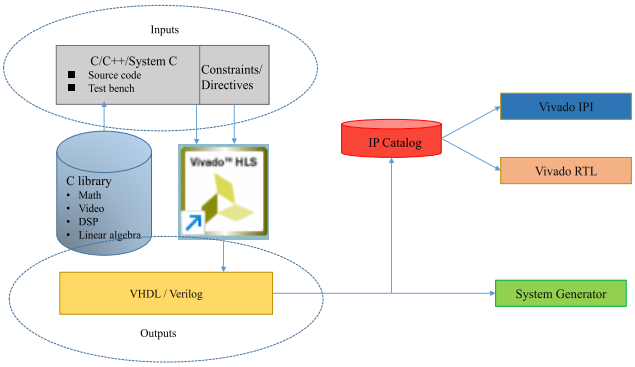


Fig. 4. Flow of HLS.

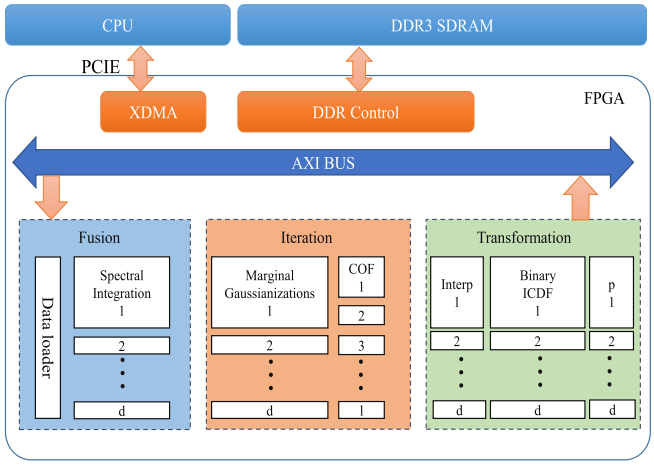


Fig. 5. Overall hardware structure of ORBIG.

does FPGA have performance advantages, but it also excels in energy efficiency. Compared to CPUs and GPUs, FPGA typically offers a better energy efficiency ratio for specific tasks, helping to reduce energy consumption and operational costs, especially suitable for applications that require long-running operations.

A. Overall Hardware Architecture

FPGA allows developers to design custom hardware logic, achieving fine control and optimization of algorithms while also providing hardware-level security features to enhance the confidentiality of data and algorithms [31]. This article considers using high-level synthesis (HLS) for FPGA engineering. HLS allows developers to use high-level programming languages such as C/C++ to develop FPGA hardware logic, greatly simplifying the FPGA design and development process. Compared to the traditional FPGA development process based on hardware description language (HDL), HLS accelerates the process from concept to hardware implementation by providing an experience closer to software development. Developers can validate the functionality of modules in a software environment. Then the HLS tool automatically converts high-level language code into HDL, seamlessly integrating the hardware simulation environment to generate modules that can be directly implemented on FPGA. Fig. 4 shows the HLS flowchart for FPGA development using Vivado HLS. Inputs include C/C++ or System C source code, a testbench, and constraints and directives to guide the synthesis process. It also includes C libraries for mathematics, video processing, digital signal processing, and linear algebra. The core processing tool is Vivado HLS, which converts high-level source code into synthesizable VHDL or Verilog code. The output is VHDL or Verilog HDL code, which can be connected to other tools, such as IP catalog, Vivado RTL, and system generator.

Fig. 5 illustrates the overall architecture of the algorithm, the implementation of the ORBIG algorithm on FPGA focuses on data flow and resource utilization. Due to the vast amount of HSI, it needs to be stored in external DDR3 SDRAM. In the spectral fusion module, we use a simultaneous read-and-process approach to avoid the need for massive data reads, where each band (band length/target dimensionality) is read in each iteration, followed by immediate fusion processing. The fusion process between different pixels lacks correlation, so this module can be optimized through PIPELINE instructions. Additionally, edge Gaussianization and linear rotation are applied to the dimensionality-reduced data. This article proposes a correlation-based rotation method, acquiring rotation matrices independent of spectral data. The rotation process can be parallelized sample-wise, while edge Gaussianization is implemented as a task-level pipelining in the spectral dimension. Therefore, the entire process achieves high throughput data processing through various levels of parallelization. Finally, the density distribution of the data in the original domain is obtained through transformation matrices and HSIs. This process involves significant matrix calculations; thus, in cases where BRAM resource consumption is controllable, array segmentation is performed. In HLS, arrays are mapped to FPGA's BRAM resources. BRAM in FPGA can only perform one read/write operation per clock cycle. Multiple dispersed BRAM resources are used instead of a single large one to parallelize the reading and writing of multiple datasets. The probability density estimation of the original image mainly operates in the spectral dimension. Thus, the overall transformation operation can be divided. In such cases, the ARRAY PARTITION parameter provided by Vivado HLS can be utilized to partition arrays flexibly and parallelize operations without data dependency. In addition to PIPELINE instructions, UNROLL is used in actual calculations for low-dimensional computations, enabling processing with the maximum loop iterations parallelism and thus improving computational speed.

B. Bisection-Based Normal Inverse CDF

In RBIG, a critical computational step involves the ICDF of the standard normal distribution. Specifically, it is necessary to calculate and return the quantiles of the standard normal distribution based on probability values. In many programming environments and libraries, such as MATLAB, Excel, and Python, similar functions like "norminv" can be used directly. However, there are no functions available for direct use, one can use lookup tables (LUTs) to implement this method in

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Algorithm 2 Bisection-Based Norminv

Require: p, low $= -10.0$, high $= 10.0$, threshold δ
1: **if** $p \leq 0.0$ **or** $p \geq 1.0$ **then**
2: **throw** Exception("Parameter p must be in the interval (0, 1).")
3: **end if**
4: **while** high $-$ low $> \delta$ **do**
5: mid $=$ (low $+$ high)$/2.0$
6: $\phi(\text{mid}) = 0.5 \times (1.0 + \text{erf}(\text{mid}/\sqrt{2.0}))$
7: **if** $\phi(\text{mid}) < p$ **then**
8: low $=$ mid
9: **else**
10: high $=$ mid
11: **end if**
12: **end while**
13: **return** mean $+$ mid $\times$ stddev

FPGA. For complex logic functions with multiple inputs, the required storage space could become large, potentially leading to resource wastage. Therefore, a simpler approach is employed to accomplish this computation. The process of the algorithm is in Algorithm 2.

The binary search method has significant advantages when implementing the normal inverse cdf algorithm on FPGA. First, the algorithm logic of binary search is simple and intuitive, facilitating implementation on resource-constrained FPGA hardware and ensuring the algorithm's execution time is highly predictable, crucial for systems requiring real-time processing. Second, since binary search does not require storing many intermediate calculation results during execution, it is more economical regarding storage resource consumption than complex numerical methods. Furthermore, the efficiency of binary search allows it to fully utilize the parallel processing capabilities of FPGA, especially when performing simple comparison and arithmetic operations. Finally, the simplicity of binary search means it is easy to optimize for specific hardware platforms, such as by adjusting computational precision or improving data paths for better performance.

VI. Experiments

In this section, we conducted experiments on real hyperspectral datasets to demonstrate the reliable performance of the proposed method. The content mainly includes an introduction to the dataset, verification of the effects of the method proposed in this article on the dataset, and a comparison of algorithm execution times after parallel optimization. The computer used in this study is equipped with an Intel[1] Core[2] i7-9700K CPU @ 3.6 GHz processor, and the FPGA is a Zynq-7000 XILINX development board, model AX7Z035B. It integrates a Zynq7000 series chip, specifically the XC7Z035-2FFG676 model, which combines the capabilities of an ARM processing system with FPGA programmable logic (PL). The dual arrangement of DDR3 memory on both the processor system (PS) and PL sides allows for efficient data processing and storage, with each side having access to 1 GB of memory (2 $\times$ 512 M bytes). The 8 GB eMMC FLASH and 512 Mb QSPI FLASH provide ample space for the operating system, file system, and user data, ensuring a robust platform for various applications.

[1]Registered Trademark.
[2]Trademarked.

TABLE I
Image Attributes Used for the Experiments

Images	Sensor	Size	Bands	SR[m]
Cat-Island	AVIRIS	150×150	188	17.2
Texas-Coast	AVIRIS	100×100	204	17.2
WTC	AVIRIS	200×200	224	1.7

A. Datasets

Table I details the main characteristics of different datasets, including key information such as the type of sensors used, spatial resolution, and image size. Specifically, the Cat-Island dataset contains images captured from an airplane flying over a beach area. In this image, a tiny white spot is observed in the center of the beach, which appears distinctly different from the surrounding environment and is thus marked as an anomaly. This anomaly is very small in proportion, occupying only 0.09% of the total pixels in the image. In contrast, the AD results of the Texas-Coast dataset indicate that the anomalous areas occupy 0.67% of the scene. The anomalous features captured in the images of this dataset, including building roofs covered by trees and bright spots caused by reflected light, are all identified as anomalies in data processing. AVIRIS collected World Trade Center (WTC) images in the New York area. The dataset covers potential hotspots corresponding to fires in the WTC, occupying 0.23% of the entire image.

B. Target Detection Accuracy Evaluation

We conduct tests in simulated and real scenarios with different dimensions and sample sizes. Specifically, we quantitatively evaluate the detection capability using the area under the curve (AUC) scoring method. This section compares the AUC results between our proposed method, ORBIG, and the traditional RBIG method. In addition to the original images, we also provide detection visualizations for each method to assess their quality through visual inspection.

In this study, the traditional RBIG method employs PCA for feature extraction and utilizes RND as its core rotation mechanism. To visually demonstrate the effects of the two methods, we perform visual comparisons among the original HSIs, the results processed by RBIG, and the results processed using the optimization methods proposed in this article.

In the experiment, we compared our proposed method with the basic method RBIG, the classical algorithm RX, and two newly proposed methods, LEBSR [32] and IEEPST [33]. Fig. 6 presents the detection outcomes for three datasets: Cat-Island, Texas-Coast, and WTC, from top to bottom, respectively. The left column displays the actual scenes of the utilized data, while the subsequent columns illustrate the detection results obtained using different algorithms. For the Cat-Island dataset, the proposed ORBIG algorithm effectively identifies anomalies within the images. In contrast, the RX, LEBSR, and IEEPST algorithms perform well in suppressing

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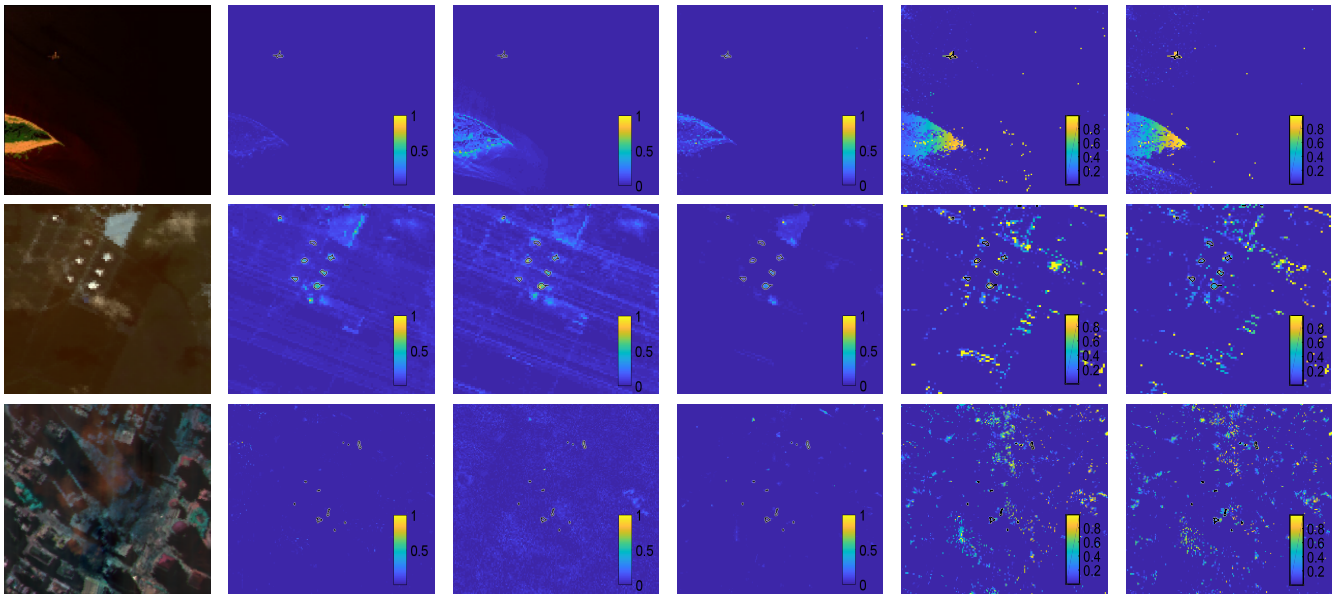


Fig. 6. AD predictions in experimental images. (a) Input. (b) RX. (c) LEBSR. (d) IEEPST. (e) RBIG. (f) ORBIG.

background but fail to highlight the anomalies, making it challenging to confirm the presence of anomalies. In the Texas-Coast dataset, background pixel interference significantly affects all algorithms. RX, LEBSR, and IEEPST algorithms detect anomalies with relatively weak signals. Although the IEEPST algorithm achieves the best background suppression, it also results in the least complete AD. Conversely, the proposed ORBIG algorithm successfully identifies anomalies with higher anomaly scores, indicating a more robust detection capability. For the WTC dataset, the RX algorithm is unable to distinguish anomalies from the background. The LEBSR algorithm mixes anomalies with the background, making it difficult to pinpoint the exact locations of anomalies. The IEEPST algorithm identifies the positions of anomalies and achieves the best background suppression, but its AD is incomplete. The proposed ORBIG algorithm, despite being significantly affected by background interference in this dataset, can still identify the positions of anomalous pixels with reasonable accuracy. These observations highlight the effectiveness of the ORBIG algorithm in detecting anomalies across different datasets, demonstrating its potential advantages in terms of computational efficiency and adaptability to specific hardware architectures, as further explored in the context of FPGA optimization in this article.

By analyzing the performance scores of these methods in Table II, we observe that the ORBIG method consistently demonstrates superior performance across the evaluated datasets, except for the Texas-Coast dataset. In the Cat-Island and WTC datasets, ORBIG achieves the highest AUC values of 0.9923 and 0.9732, respectively, indicating its strong capability in detecting anomalies in these scenarios. However, in the Texas-Coast dataset, the IEEPST method stands out with the highest AUC score, outperforming ORBIG and other methods. This superior performance of IEEPST on the Texas-Coast dataset is attributed to its effectiveness on datasets with a higher proportion of anomalies, a factor that the ORBIG method has not yet fully addressed but plans to improve in future work. Overall, the comparative analysis showcases the effectiveness of the ORBIG method in enhancing detection performance, confirming its potential for practical applications in hyperspectral imaging.

TABLE II
AUC VALUES OF THE COMPARABLE METHODS FOR THREE DATASETS

Images	RX	LEBSR	IEEPST	RBIG	ORBIG
Cat-Island	0.9612	0.9701	0.9833	0.9917	0.9923
Texas-Coast	0.9310	0.9485	0.9987	0.9391	0.9538
WTC	0.9501	0.8967	0.9569	0.9603	0.9732

C. Computational Efficiency

The data presented in Table III reveals a significant impact of different linear rotation methods on the convergence speed of the algorithm when processing HSIs of Cat-Island, Texas-Coast, and WTC. Using the traditional RBIG method, processing the Cat-Island image required 198 iterations of marginalization and rotation operations for algorithm convergence, whereas employing the correlation-based rotation method proposed in this article significantly reduced the convergence iterations to 109. Similarly, for the Texas-Coast dataset, the correlation-based rotation method demonstrated higher efficiency than RBIG, reducing the iteration count from 189 to 133. In the case of the WTC dataset, the ORBIG algorithm reduced the convergence iterations from 201 to 110. These results demonstrate that the method proposed in this study significantly contributes to accelerating the convergence speed of the RBIG algorithm, effectively enhancing the efficiency of HSI processing.

Further analysis reveals that the reduction in iteration count directly affects the scale of the transformation matrix,

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TABLE III
NUMBER OF CONVERGENCE ITERATIONS

Images	RBIG	ORBIG
Cat-Island	198	109
Texas-Coast	189	133
WTC	201	110

TABLE IV
TIME CONSUMPTION UNDER DIFFERENT ENVIRONMENTS

	ORBIG(MATLAB)	ORBIG(C++)	ORBIG(FPGA)
Cat-Island	97.56s	647.26s	12.34s
Texas-Coast	52.37s	313.16s	7.10s
WTC	161.68s	948.52s	19.92s

TABLE V
TIME CONSUMPTION UNDER DIFFERENT SIZE

	origin	4 times	8 times	12 times
Time	12.34s	25.55s	51.64s	78.38s

accelerating the algorithm's convergence speed and simplifying the calculation process of restoring sample data from the transformed domain to the original domain. Such efficiency improvement is crucial in HSI processing, as it typically involves large volumes of data and complex computations. The method proposed in this article can reduce computation time and resource consumption by optimizing the iteration process, providing a new solution for the rapid and effective processing of rotation operations. Table IV presents the execution time of our proposed algorithm under different processing platforms and programming environments, including MATLAB, C++, and FPGA architecture. We can analyze the algorithm's time efficiency and adaptability on various hardware platforms through these data, thus providing robust, practical guidance for HSI target detection.

In the MATLAB environment, the processing time for the Cat-Island dataset was 97.56 and 52.37 s for the Texas-Coast dataset, and 161.68 s for the larger WTC dataset. In contrast, the processing time significantly increased when the algorithm was implemented in the C++ environment. It rose to 647.26 s for the Cat-Island dataset, 313.16 s for the Texas-Coast dataset, and 948.52 s for WTC. This significant time difference may partly be attributed to MATLAB's built-in optimization mechanisms and the potential lack of advanced optimizations in the C++ environment. However, when deployed on the FPGA architecture, the algorithm's performance improved significantly. On the FPGA platform, the processing time for the Cat-Island dataset was only 12.34 s, reduced to 7.10 s for the Texas-Coast dataset and 19.92 s for WTC. This result highlights the significant potential of FPGA as a parallel execution hardware facility in enhancing code execution efficiency. Particularly in computationally intensive tasks such as HSI target detection, applying FPGA can significantly enhance algorithmic efficiency, providing an attractive solution for real-time HSI analysis. However, it is worth noting that despite FPGA's advantages in time efficiency, it may still encounter limitations in terms of storage and computational resources when processing large-sized or high-bandwidth HSIs.

Table V illustrates the time required by the ORBIG algorithm to process datasets of varying sizes. To better understand the scalability and efficiency of our method, we took the Cat-Island dataset as a representative example. We systematically expanded the dataset by multiples and meticulously measured the corresponding processing times. The results reveal that there is a clear positive correlation between the data size and the processing time. However, it's important to note that this relationship is not a straightforward linear or exact multiple correlation. As the volume of data increases, several additional factors come into play that affect the overall processing time. Specifically, as the dataset grows larger, the computational resources, particularly in FPGA implementations, face increased competition for memory and processing units. This leads to additional waiting times and potential bottlenecks during various stages of computation, such as data transfer, parallel processing synchronization, and memory access conflicts. These factors collectively contribute to longer processing times, highlighting the complexity and challenges of scaling up the data processing capabilities.

VII. CONCLUSION

This article successfully proposes and validates an FPGA-based hyperspectral AD algorithm to address the challenge of low efficiency in HSI processing. By fully leveraging FPGA's unique advantages as a hardware computing platform, this research improves throughput and computational efficiency by introducing strategies such as the averaging fusion algorithm, optimizing the Gaussianization process, and binary search. These methods fully demonstrate FPGA's strong capabilities in parallel processing and precise computation.

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